

SIH26058

HARDWARE EDITION

Robotics and Drones

Challenge #58 of 236

Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs)

Sponsoring Organization: Ministry of Earth Sciences (MoES)

Department: National Institute of Ocean Technology (NIOT)

Theme / Domain: Robotics and Drones

Submission Deadline: **30 September 2026**

Submitted Ideas: 0/500

Cash Prize: ₹1,00,000 / Winning Team

OFFICIAL PROBLEM STATEMENT BACKGROUND & REQUIREMENTS:

- Background In underwater exploration and marine mapping, Autonomous Underwater Vehicles (AUVs) rely heavily on side-scan sonar systems. The performance of these systems is entirely dependent on the physical characteristics of the transmitted acoustic wave, known as the 'ping'. Traditional sonars transmit short, fixed-frequency pulses. However, modern advanced military and research systems utilize Linear Frequency Modulated (LFM) Chirps "waveforms that sweep across a spectrum of frequencies over a precise timeframe. The primary bottleneck is that the underwater environment is highly dynamic. Sound waves behave differently depending on water depth, turbidity (suspended mud/sediment particles), temperature, and salinity. A high-frequency chirp (500 kHz) offers ultra-high image resolution but scatters instantly in muddy or deep waters. Conversely, a low-frequency chirp (100 kHz) can penetrate murky water and travel long distances but yields a blurry, low-resolution image. For an AUV to map effectively without draining its limited battery payload, its transmitter hardware must behave like a Software-Defined Radio (SDR) "dynamically adapting its physical analog pulse waveform in real-time based on the actual environmental conditions it encounters.
- Description Participants must design, prototype, and demonstrate a physical, self-contained Software-Defined Sonar Transmitter Payload Module. Instead of a software simulation, the solution must be a physical hardware unit built using an embedded platform (e.g., STM32, ESP32, Texas Instruments DSP, or an FPGA) integrated with custom analog electronics. The hardware must ingest real-time environmental data (via physical sensors, or analog voltage dials acting as sensor inputs) and mathematically synthesize and output an optimized, real-time physical analog waveform via a Digital-to-Analog Converter (DAC) and amplifier circuit. The entire hardware architecture must focus heavily on low-power consumption and hardware level optimization. Teams must utilize low-level configurations (such as Direct Memory Access (DMA) and hardware timers) to ensure the processing unit does not drain a marine drone's battery pack while trying to compute complex trigonometric wave values under strict real-time constraints.
- Expected Solution Teams are expected to deliver a functional physical hardware prototype consisting of the following modules:
- Embedded Firmware Engine: A robust program deployed on a physical microcontroller or FPGA (written in C/C++, Verilog, or VHDL). The firmware must utilize hardware timers and DMA to stream wave-generation arrays directly to an internal or external DAC without stalling the CPU. The system must support multiple modulation types on the fly: LFM chirps, geometric sweeps, and phase-coded pulses.
- Environmental Sensor Interface & Adaptation Logic: A physical control interface where real-time environmental changes are introduced to the hardware (via physical sensors, or potentiometers simulating sensors for parameters like 'Entering Muddy Estuary' or 'Entering Clear Shallow Reef'). The microcontroller must read these inputs via an ADC and modify three critical wave parameters instantly: 1. Bandwidth/Center Frequency (Tuning for range vs. resolution) 2. Pulse Duration (Controlling total energy output) 3. Amplitude/Signal Power
- Analog Signal Conditioning & Hardware Filters: A physical analog frontend circuit (built on a breadboard or custom PCB) featuring active/passive low-pass filters and an operational amplifier. Combined with digital windowing filters (such as Hamming, Hann, or Blackman windows) applied in the firmware, this hardware must smooth out sudden voltage jumps at the start and end of a pulse, protecting the transmitter hardware from electrical stress and eliminating sidelobe artifacts.
- Physical Form Factor & Output Validation: The physical analog output of the transmitter payload must be connected to an oscilloscope or spectrum analyzer at the judging table. The generated raw waves must demonstrate clean, low-distortion, mathematically sound spectrograms when validated via a Fast Fourier Transform (FFT). Additionally, the module should be housed in a robust, 3D-printed or fabricated structural enclosure representing a field-deployable payload pod designed for an AUV hull slot.